

DARIL Functional Requirements

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Introduction

This project involves the development of an **application** through which DQCE members, AIT managers, and Reporting teams can view and better understand the lineage of data as it flows through the Bank’s network. Currently, access to this lineage data is difficult to obtain and requires the user to go through many tedious and redundant steps. However, because this data is stored within the bank’s network and is presumably accessible upon request, this means that we can pre-calculate lineage and stitch together a picture for users to access when required. This would reduce the need to trace lineage back manually as this could be seen in a single picture.

Our vision of a final product involves a **2D network visualization** that allows the user¹ to see how **data, applications, and reports** within the **bank’s network** are **connected**. These pictures would be pre-calculated and available on demand making the process much smoother than was prior. The primary objective for the application would be to provide data understanding for a selected ADS or ERR Report. Therefore, the user will select a specific ADS or Report before being presented with a subset of the Bank’s network that is associated with said entry. It is important to note that because the Bank’s network is a size beyond comprehension, this subset is required to provide the user with a smooth viewing experience.

Also, the user can further explore the selected subset by highlighting certain associations. Associations will include but not limited to:

- The path of a **PDE** through the network
- The path of all **PDEs directly related to a selected AIT** through the network.
- The path of all data elements related to a **selected report** through the network.
- The path of all data elements related to all reports associated with a **selected regulator**.

At any point the user may reset the visualization to return to the original subset or select a new one.

1. See system roles below for more detail.

Roles:

- **User:** Any member of the DQCE, AIT, Business, or Reporting teams that use the application to visualize data flow.

Requirements

1. The **system** shall prompt the user upon visiting the application to select an ADS or Report they would like to see a subset of the Bank’s network for.

#	User Story
1	The prompt may be switched between showing ADS' loaded into the system or Reports loaded into the system.
2	The prompt shall be a dropdown list with a set number of entries to choose from
3	The prompt shall default to display ADS options
4	The prompt shall default to a blank value
5	The user may change the input at any time

2. The **system** shall display a network visualization modeling the Bank's network of AITs, PDEs, Reports, and Other sources in a selected subset

#	User Story
1	The system may display up to 1 hops removed connections upstream from the source ADS
2	The system shall distinguish PDEs, AITs, Reports, and Other sources from each other
3	The system shall distinguish ADS' from AITs
4	The system shall distinguish AITs and Reports with DART incidents from those that do not
5	The system shall differentiate network edges based on usage

3. A **user** may choose to interact with the model highlighting important information in the subset

#	User Story
1	The user may select a specific PDE to highlight its flow through the network
2	The user may select a specific Report to highlight its associated PDEs through the network.
3	The user may select a specific Regulator to see PDEs associated with all Reports linked to the selected Regulator and how they flow through the network.
4	The user may select a specific AIT to see how all PDE's directly associated with the selected application flow through the network.
5	The user's selections from 3.1-3.4 may compound on top of each other.
6	The user may reset their selections at any time.
7	The user may hover over a Report node to see it's percent of data is coming from associated source ADS(s) in the view].
8	The user may hover over an AIT to see its related data and up to 5 associated DARTs
9	The user may hover over a PDE to see its related data
10	The user may hover over an 'Other' source to see more detail

4. The **system** may filter to a "system" mode.

#	User Story
1	The user may toggle between System and PDE mode.

2	The system in PDE mode displays the existing network
3	The system in system mode displays the existing network without associated PDEs
4	The user's choice is persisted until toggled back to PDE mode

5. The **user** may export the visualization data

#	User Story
1	Data shall include mappings between Reports and their associated PDEs and the AITs the PDEs flow through
2	Data shall include metadata related to Reports in the view
3	Data shall reflect any filters selected by the user

Non-Functional Requirements:

1. No **user's** session should be persisted from one session to the next
2. No **user's** session should affect a different user's session.
3. The **system** shall be refreshed on a **weekly**
4. The **system's** extract files (report to source) shall be refreshed on a **quarterly**
5. The model should be pre-calculated for the user upon arrival.
6. The visualization should be intuitive and easy to use.
7. The application should be web hosted.

Constraints:

1. The network visualization model shall use force directed network principles.
2. Exported data shall be in a XLSX format.

requirements